CAS SciFinder

Task History

Initiating Search

November 4, 2025, 11:31 AM

Search:

CAS Lexicon Search:

Stille coupling reaction - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (4,657)	References	View Results
Exported: Retrieved Related Reaction Results + Filters (3,197) Filtered By: Yield: 90-100%, 80-89% Language: English	Reactions	View Results

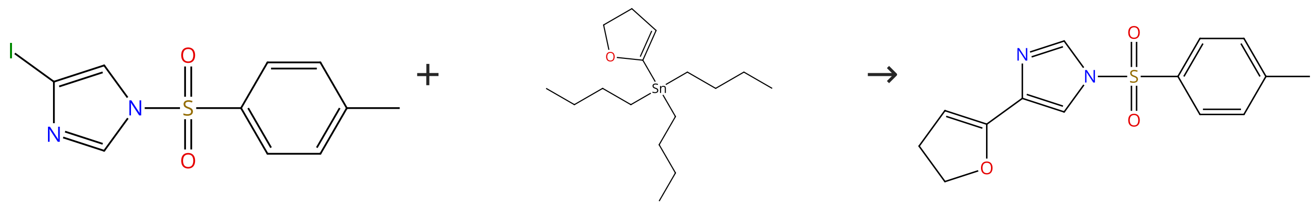
CAS SciFinder

Reactions (50)

View in CAS SciFinder

Scheme 1 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (36)

Suppliers (25)

31-130-CAS-9716226

Steps: **1** Yield: **100%**

1.1 **Catalysts:** Tetrakis(triphenylphosphine)palladium
Solvents: Dimethylformamide; 2 h, 110 °C

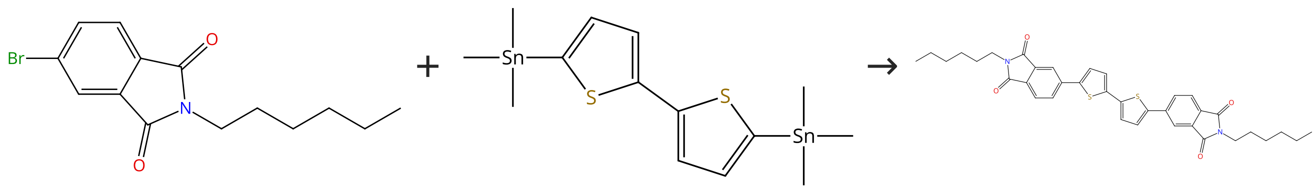
Stille Cross-Coupling for the Functionalization of the Imidazole Backbone: Revisit, Improvement, and Applications of the Method

By: Sandtorv, Alexander H.; et al

European Journal of Organic Chemistry (2015), 2015(16), 3506-3512.

Scheme 2 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (3)

Suppliers (53)

31-130-CAS-15803527

Steps: **1** Yield: **100%**

1.1 **Solvents:** Toluene; 5 min, 100 °C; 20 min, 170 °C

Experimental Protocols

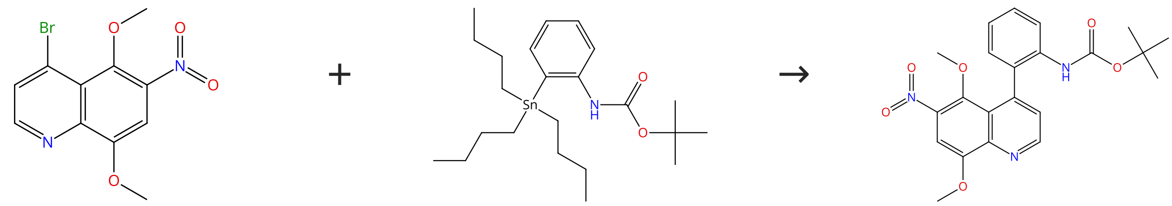
Utility of a heterogeneous palladium catalyst for the synthesis of a molecular semiconductor via Stille, Suzuki, and direct heteroarylation cross-coupling reactions

By: McAfee, Seth M.; et al

RSC Advances (2015), 5(33), 26097-26106.

Scheme 3 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (2)

Suppliers (2)

31-130-CAS-68496 Steps: **1** Yield: **100%**

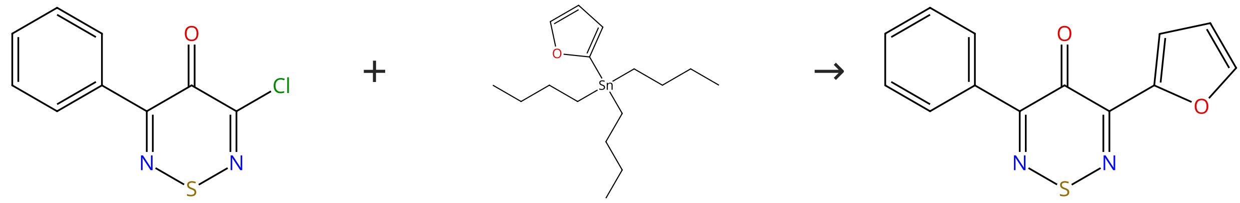
1.1 **Catalysts:** Palladium, bis(acetonitrile)dichloro-
Solvents: 1,4-Dioxane

A C-Ring regioisomer of the marine alkaloid meridine exhibits selective in vitro cytotoxicity for solid tumours

By: Angel de la Fuente, J.; et al

Bioorganic & Medicinal Chemistry (2001), 9(7), 1807-1814.

Scheme 4 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (2) | Suppliers (45) | Supplier (1)

31-130-CAS-9354993 Steps: **1** Yield: **100%**

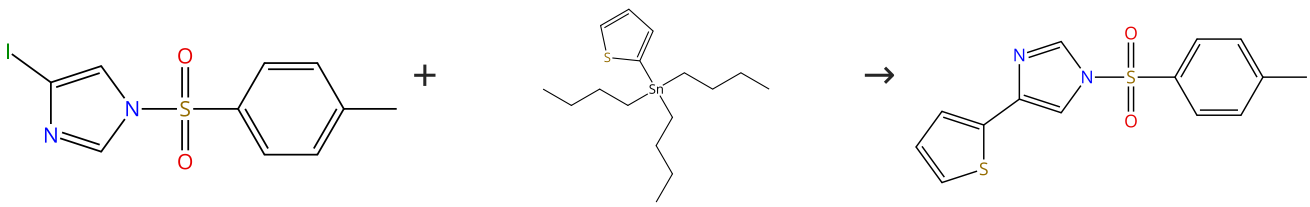
1.1 **Catalysts:** Dichlorobis(triphenylphosphine)palladium
Solvents: Acetonitrile; 20 °C; 3.8 h, 82 °C

Synthesis of asymmetric 3,5-diaryl-4H-1,2,6-thiadiazin-4-ones via Suzuki-Miyaura and Stille coupling reactions

By: Ioannidou, Heraklidia A.; et al

Tetrahedron (2012), 68(36), 7380-7385.

Scheme 5 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (36) | Suppliers (38)

31-130-CAS-14308511 Steps: **1** Yield: **100%**

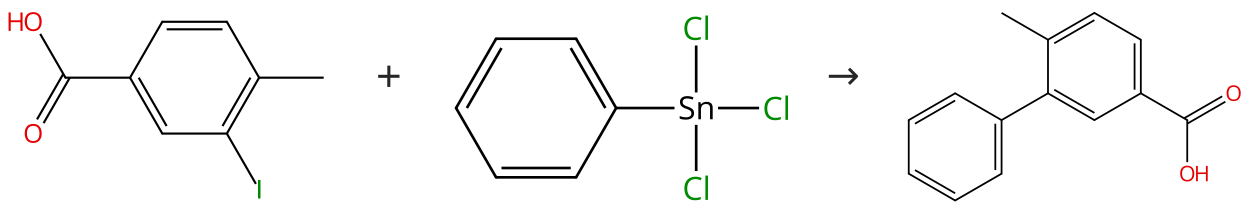
1.1 **Catalysts:** Cuprous iodide, Tetrakis(triphenylphosphine) palladium
Solvents: Dimethylformamide; 2 h, 110 °C

Stille Cross-Coupling for the Functionalization of the Imidazole Backbone: Revisit, Improvement, and Applications of the Method

By: Sandtorv, Alexander H.; et al

European Journal of Organic Chemistry (2015), 2015(16), 3506-3512.

Scheme 6 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (90) | Suppliers (51) | Suppliers (35)

31-130-CAS-1314477 Steps: **1** Yield: **100%**

1.1 **Reagents:** Potassium hydroxide
Catalysts: Palladium
Solvents: Water; rt

1.2 **Reagents:** Hydrochloric acid
Solvents: Water; rt

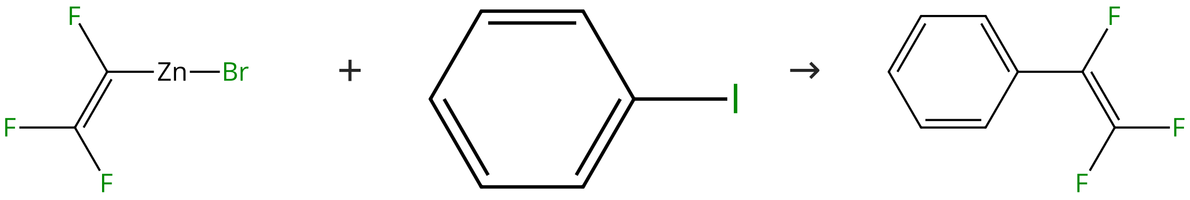
Experimental Protocols

Dendrimer-Encapsulated Pd Nanoparticles as Aqueous, Room-Temperature Catalysts for the Stille Reaction

By: Garcia-Martinez, Joaquin C.; et al

Journal of the American Chemical Society (2005), 127(14), 5097-5103.

Scheme 7 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (2) Suppliers (95) Suppliers (21)

31-176-CAS-6679654 Steps: **1** Yield: **100%**

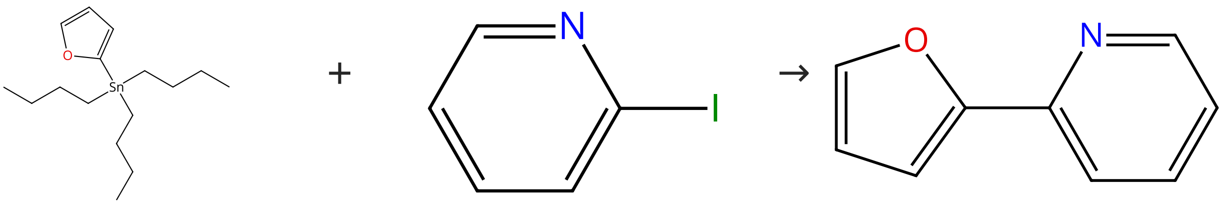
1.1 **Catalysts:** Tetrakis(triphenylphosphine)palladium
Solvents: Dimethylformamide; 12 h, rt

The stereoselective synthesis of (Z)-HFC=CFZnI and stereospecific preparation of (E)-1,2-difluorostyrenes from (Z)-HFC=CFZnI via an unusual Pd(PPh_3)$_4$-Cu(I)Br co-catalysis approach or (Z)-HFC=CFSnBu$_3$

By: Liu, Qibo; et al

Journal of Fluorine Chemistry (2011), 132(2), 78-87.

Scheme 8 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (45) Suppliers (79) Suppliers (32)

31-130-CAS-256153 Steps: **1** Yield: **100%**

1.1 **Reagents:** Lithium chloride
Catalysts: Dichlorobis(triphenylphosphine)palladium
Solvents: Tetrahydrofuran; 480 min, 85 °C

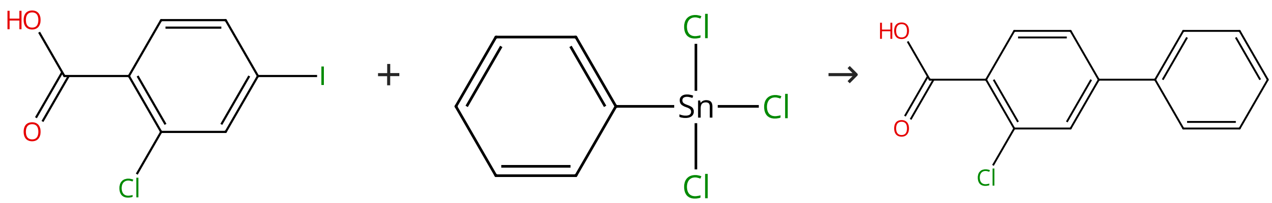
Experimental Protocols

Dual Effect of Halides in the Stille Reaction: In Situ Halide Metathesis and Catalyst Stabilization

By: Verbeeck, Stefan; et al

Chemistry - A European Journal (2010), 16(43), 12831-12837, S12831/1-S12831/17.

Scheme 9 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (67) Suppliers (51) Suppliers (49)

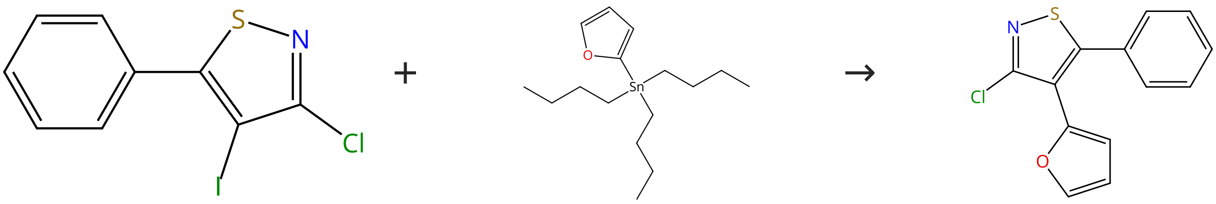
31-130-CAS-13310233	Steps: **1** Yield: **100%**
1.1	**Reagents:** Potassium hydroxide **Catalysts:** Palladium, L-Threonyl-L-seryl-L-asparaginyl-L-alanyl-L-valyl-L-histidyl-L-prolyl-L-threonyl-L-leucyl-L-arginyl-L-histidyl-L-leucine **Solvents:** Water; 24 h, 40 °C
1.2	**Reagents:** Hydrochloric acid **Solvents:** Water
Experimental Protocols	

Exploring the mechanism of Stille C-C coupling via peptide-capped Pd nanoparticles results in low temperature reagent selectivity

By: Pacardo, Dennis B.; et al

Catalysis Science & Technology (2013), 3(3), 745-753.

Scheme 10 (1 Reaction) Steps: **1** Yield: **100%**



Supplier (1) Suppliers (45) Supplier (1)

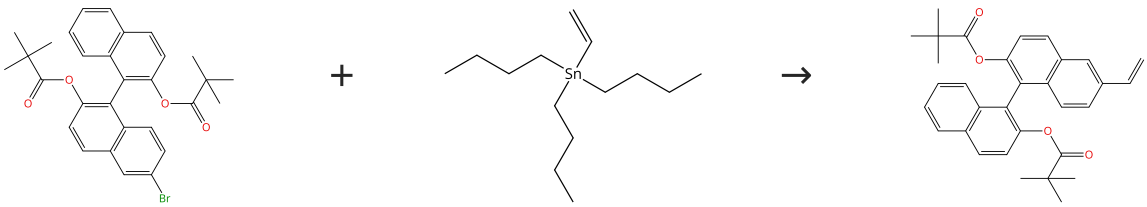
31-130-CAS-392546	Steps: **1** Yield: **100%**
1.1	**Catalysts:** Palladium diacetate **Solvents:** Dimethylformamide; rt → 100 °C; 25 min, 100 °C; 100 °C → 20 °C

3,4,5-Triarylisothiazoles via C-C coupling chemistry

By: Christoforou, Irene C.; et al

Organic & Biomolecular Chemistry (2007), 5(9), 1381-1390.

Scheme 11 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (61)

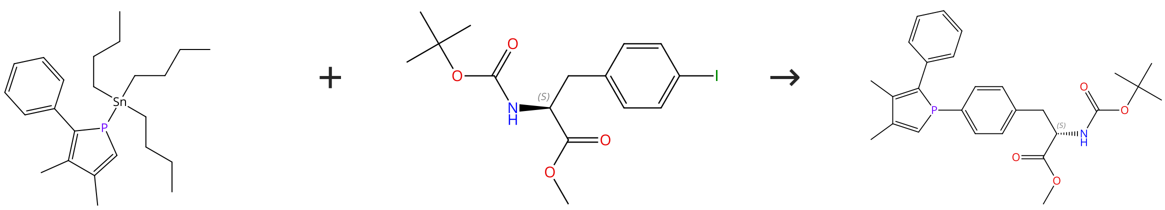
31-130-CAS-9785363	Steps: **1** Yield: **100%**
1.1	**Catalysts:** Tetrakis(triphenylphosphine)palladium **Solvents:** Dimethylformamide; 4 h, 100 °C
1.2	**Reagents:** Potassium fluoride **Solvents:** Toluene, Water; overnight, rt
Experimental Protocols	

Palladium-catalyzed reactions of 6-halogeno-1,1'-binaphthyl derivatives. A detailed investigation of structure/reactivity and structure/selectivity relationships

By: Feher, Csaba; et al

Tetrahedron (2011), 67(34), 6327-6333.

Scheme 12 (1 Reaction) Steps: **1** Yield: **100%**



Absolute stereochemistry shown, Rotation (+)

Absolute stereochemistry shown

Suppliers (52)

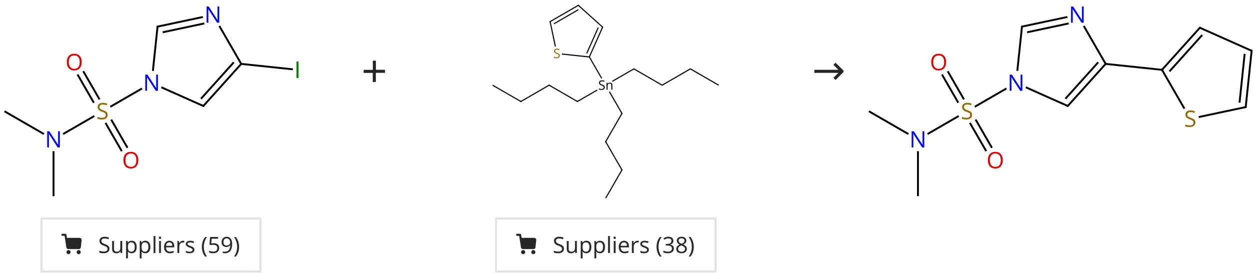
31-176-CAS-3779137	Steps: **1** Yield: **100%**
1.1	**Catalysts:** Bis(dibenzylideneacetone)palladium **Solvents:** Tetrahydrofuran; 12 h, 60 °C
Experimental Protocols	

Incorporation of phosphole moieties into the side chain of tyrosine and phenylalanine

By: Bisaro, Fabrice; et al

Synlett (2010), (20), 3081-3085.

Scheme 13 (1 Reaction) Steps: **1** Yield: **100%**



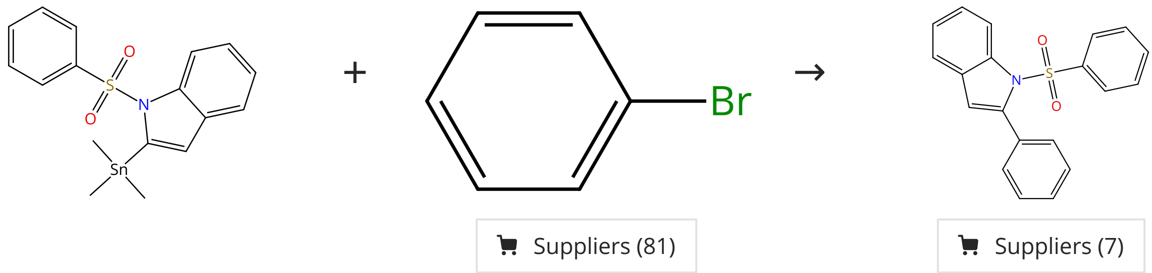
31-130-CAS-6867403	Steps: **1** Yield: **100%**
1.1	**Catalysts:** Cuprous iodide, Tetrakis(triphenylphosphine) palladium **Solvents:** Dimethylformamide; 2 h, 110 °C

Stille Cross-Coupling for the Functionalization of the Imidazole Backbone: Revisit, Improvement, and Applications of the Method

By: Sandtorv, Alexander H.; et al

European Journal of Organic Chemistry (2015), 2015(16), 3506-3512.

Scheme 14 (1 Reaction) Steps: **1** Yield: **100%**



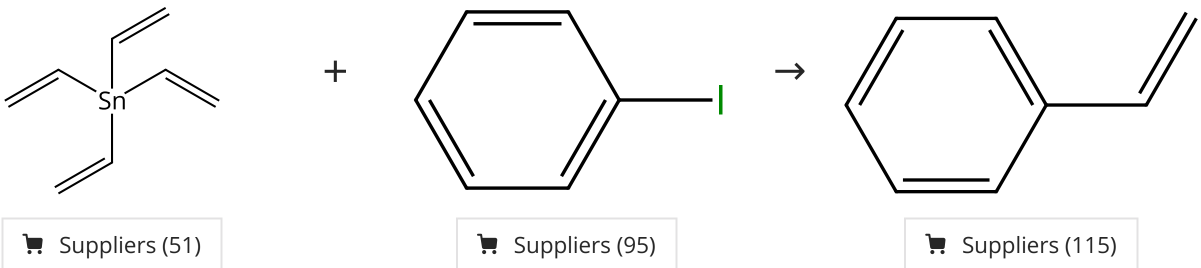
31-130-CAS-5727874	Steps: **1** Yield: **100%**
1.1	**Catalysts:** Cuprous iodide **Solvents:** Dimethylacetamide; 30 min, rt
1.2	**Catalysts:** Tetrakis(triphenylphosphine)palladium; 12 h, 100 °C

Synthesis of new fused and substituted benzo and pyrido carbazoles via C-2 (het)arylindoles

By: Bourderioux, Aurelie; et al

Tetrahedron (2008), 64(49), 11012-11019.

Scheme 15 (1 Reaction) Steps: **1** Yield: **100%**



31-130-CAS-12983133 Steps: **1** Yield: **100%**

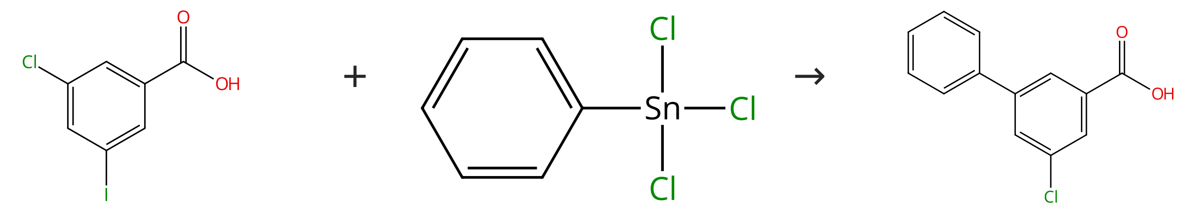
1.1 **Reagents:** Maltose, Dimethylurea, Ammonium chloride
Catalysts: Triphenylarsine, Palladium, tris[μ-[(1,2-η:4,5-η)-(1*E*,4*E*)-1,5-diphenyl-1,4-pentadien-3-one]]di-, compd. with trichloromethane (1:1); 6 h, 90 °C; 90 °C → rt

Stille reactions with tetraalkylstannanes and phenyltrialkylstannanes in low melting sugar-urea-salt mixtures

By: Imperato, Giovanni; et al

Advanced Synthesis & Catalysis (2006), 348(15), 2243-2247.

Scheme 16 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (63) Suppliers (51) Suppliers (41)

31-130-CAS-6921947 Steps: **1** Yield: **100%**

1.1 **Reagents:** Potassium hydroxide
Catalysts: Palladium, L-Threonyl-L-seryl-L-asparaginyl-L-alanyl-L-valyl-L-histidyl-L-prolyl-L-threonyl-L-leucyl-L-arginyl-L-histidyl-L-leucine
Solvents: Water; 24 h, 40 °C

1.2 **Reagents:** Hydrochloric acid
Solvents: Water

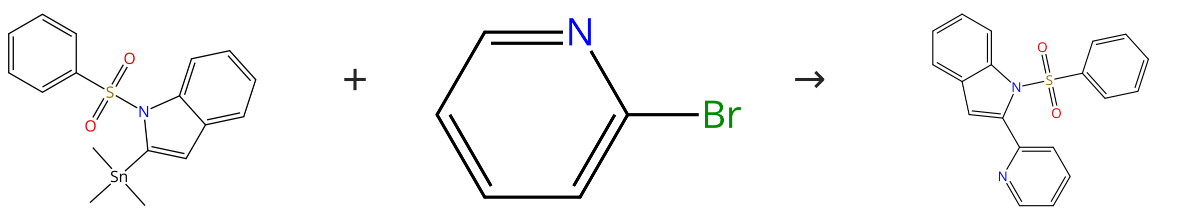
Experimental Protocols

Exploring the mechanism of Stille C-C coupling via peptide-capped Pd nanoparticles results in low temperature reagent selectivity

By: Pacardo, Dennis B.; et al

Catalysis Science & Technology (2013), 3(3), 745-753.

Scheme 17 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (87) Suppliers (4)

31-130-CAS-14860265 Steps: **1** Yield: **100%**

1.1 **Catalysts:** Cuprous iodide
Solvents: Dimethylacetamide; 30 min, rt

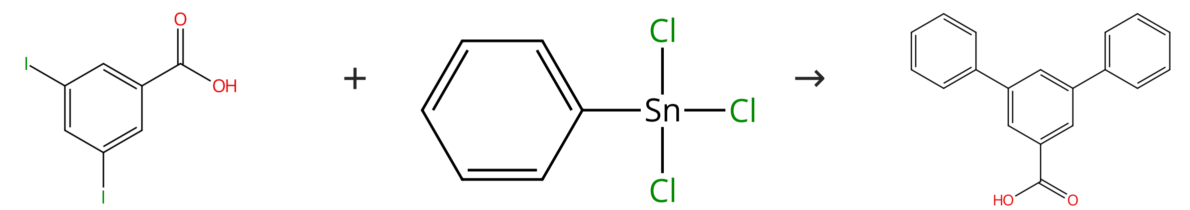
1.2 **Catalysts:** Tetrakis(triphenylphosphine)palladium; 6 h, 100 °C

Synthesis of new fused and substituted benzo and pyrido carbazoles via C-2 (het)arylindoles

By: Bourderioux, Aurelie; et al

Tetrahedron (2008), 64(49), 11012-11019.

Scheme 18 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (74) Suppliers (51) Suppliers (9)

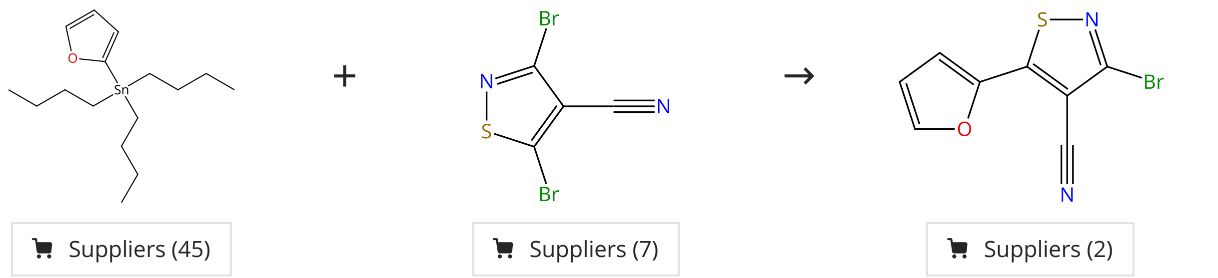
31-130-CAS-2671941	Steps: **1** Yield: **100%**
1.1	**Reagents:** Potassium hydroxide **Catalysts:** Palladium, L-Threonyl-L-seryl-L-asparaginyl-L-alanyl-L-valyl-L-histidyl-L-prolyl-L-threonyl-L-leucyl-L-arginyl-L-histidyl-L-leucine **Solvents:** Water; 24 h, 40 °C
1.2	**Reagents:** Hydrochloric acid **Solvents:** Water
Experimental Protocols	

Exploring the mechanism of Stille C-C coupling via peptide-capped Pd nanoparticles results in low temperature reagent selectivity

By: Pacardo, Dennis B.; et al

Catalysis Science & Technology (2013), 3(3), 745-753.

Scheme 19 (1 Reaction) Steps: **1** Yield: **100%**



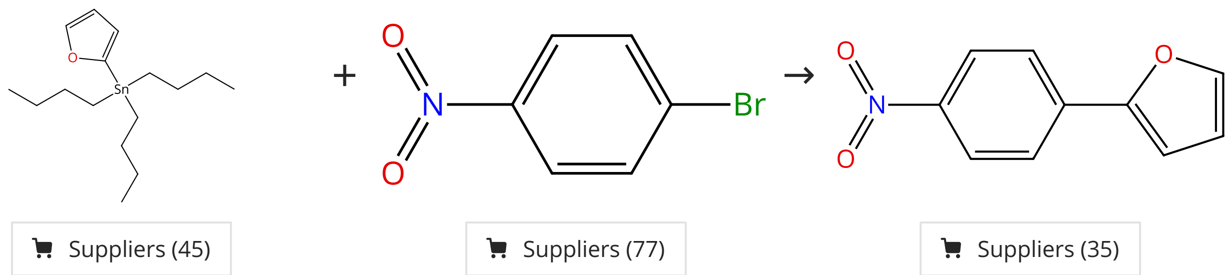
31-130-CAS-2769621	Steps: **1** Yield: **100%**
1.1	**Catalysts:** Palladium diacetate **Solvents:** Acetonitrile; 100 °C

New regiospecific isothiazole C-C coupling chemistry

By: Christoforou, Irene C.; et al

Organic & Biomolecular Chemistry (2006), 4(19), 3681-3693.

Scheme 20 (1 Reaction) Steps: **1** Yield: **100%**



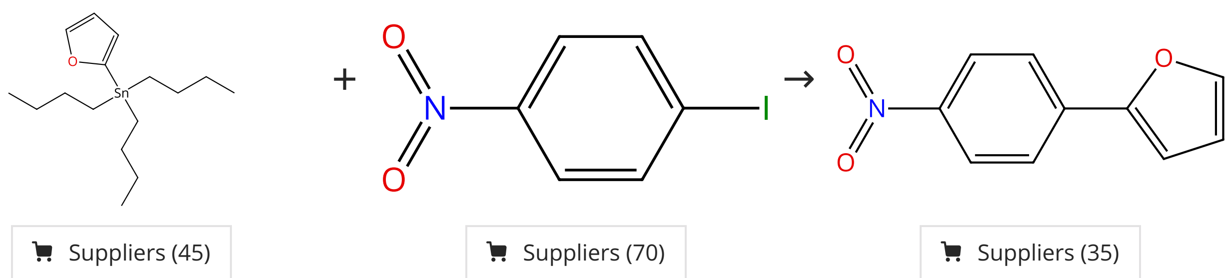
31-130-CAS-9640348	Steps: **1** Yield: **100%**
1.1	**Reagents:** Tetrabutylammonium fluoride **Catalysts:** Dabco, Palladium diacetate **Solvents:** 1,4-Dioxane; 16 h, 100 °C
Experimental Protocols	

Efficient Stille cross-coupling reaction catalyzed by the Pd(OAc)$_2$/Dabco catalytic system

By: Li, Jin-Heng; et al

Journal of Organic Chemistry (2005), 70(7), 2832-2834.

Scheme 21 (1 Reaction) Steps: **1** Yield: **100%**



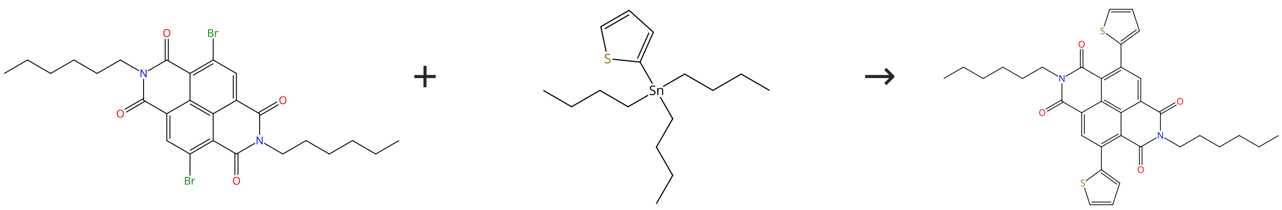
31-130-CAS-12090905	Steps: **1** Yield: **100%**
1.1	**Reagents:** Potassium fluoride **Catalysts:** *N,N'*-1,2-Ethanediylidenebis[cyclohexanamine], Bis(dibenzylideneacetone)palladium **Solvents:** 1,4-Dioxane; 16 h, 100 °C

An efficient Stille cross-coupling reaction catalyzed by Pd(dba)2/DAB-Cy catalytic system

By: Li, Jin-Heng; et al

Tetrahedron (2005), 61(30), 7289-7293.

Scheme 22 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (43) Suppliers (38)

31-130-CAS-10679561	Steps: **1** Yield: **100%**
1.1	**Catalysts:** Tri-*o*-tolylphosphine, Tris(dibenzylideneacetone)dipalladium **Solvents:** Toluene; 2 h, 90 °C
1.2	**Reagents:** Hydrochloric acid **Solvents:** Water

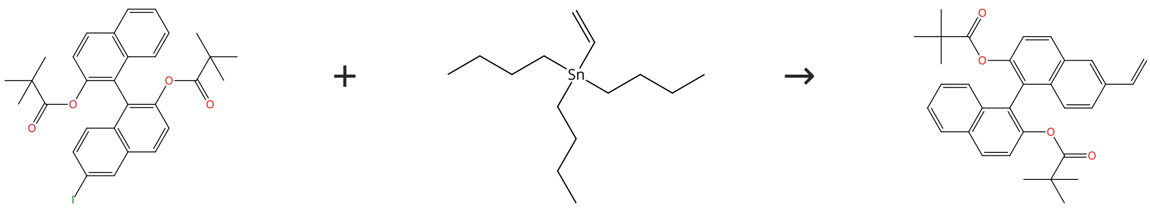
Experimental Protocols

Nonstoichiometric Stille Coupling Polycondensation for Synthesizing Naphthalene-Diimide-Based π-Conjugated Polymers

By: Goto, Eisuke; et al

ACS Macro Letters (2015), 4(9), 1004-1007.

Scheme 23 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (61)

31-130-CAS-7651920	Steps: **1** Yield: **100%**
1.1	**Catalysts:** Tetrakis(triphenylphosphine)palladium **Solvents:** Dimethylformamide; 2 h, 60 °C
1.2	**Reagents:** Potassium fluoride **Solvents:** Toluene, Water; overnight, rt

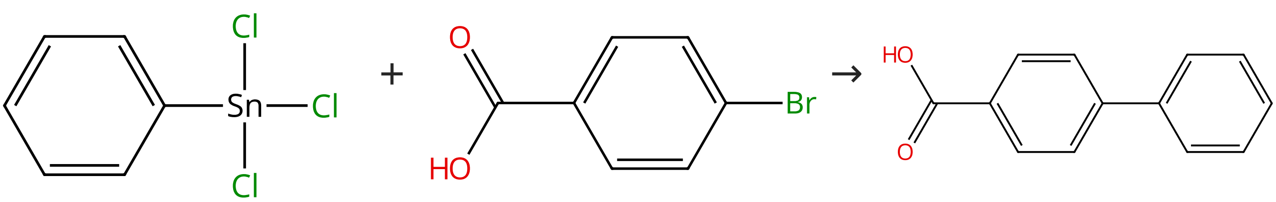
Experimental Protocols

Palladium-catalyzed reactions of 6-halogeno-1,1'-binaphthyl derivatives. A detailed investigation of structure/reactivity and structure/selectivity relationships

By: Feher, Csaba; et al

Tetrahedron (2011), 67(34), 6327-6333.

Scheme 24 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (51) Suppliers (99) Suppliers (101)

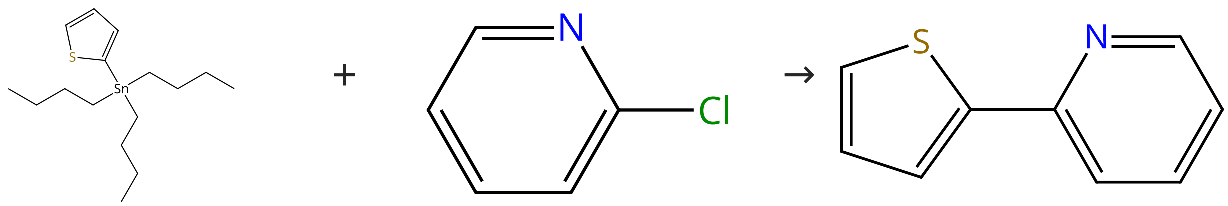
31-130-CAS-9854602	Steps: **1** Yield: **100%**
1.1	**Reagents:** Potassium hydroxide **Catalysts:** Palladium **Solvents:** Water; rt
1.2	**Reagents:** Hydrochloric acid **Solvents:** Water; rt
Experimental Protocols	

Dendrimer-Encapsulated Pd Nanoparticles as Aqueous, Room-Temperature Catalysts for the Stille Reaction

By: Garcia-Martinez, Joaquin C.; et al

Journal of the American Chemical Society (2005), 127(14), 5097-5103.

Scheme 25 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (38) Suppliers (66) Suppliers (66)

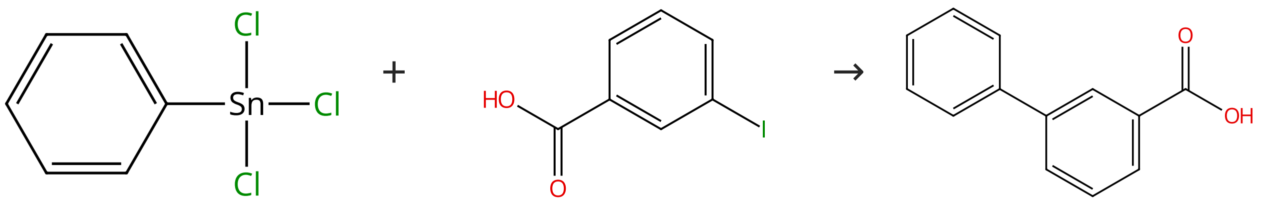
31-130-CAS-8091448	Steps: **1** Yield: **100%**
1.1	**Reagents:** Lithium chloride **Catalysts:** Dichlorobis(triphenylphosphine)palladium **Solvents:** Tetrahydrofuran; 480 min, 85 °C
Experimental Protocols	

Dual Effect of Halides in the Stille Reaction: In Situ Halide Metathesis and Catalyst Stabilization

By: Verbeeck, Stefan; et al

Chemistry - A European Journal (2010), 16(43), 12831-12837, S12831/1-S12831/17.

Scheme 26 (4 Reactions) Steps: **1** Yield: **100%**



Suppliers (51) Suppliers (86) Suppliers (88)

31-130-CAS-10534595	Steps: **1** Yield: **100%**
1.1	**Reagents:** Potassium hydroxide **Catalysts:** Palladium **Solvents:** Water; 24 h, 25 °C
1.2	**Reagents:** Hydrochloric acid **Solvents:** Water

Exploring bio-inspired strategies for the production of noble metal nanocatalysts

By: Knecht, Marc R.; et al

Polymer Preprints (American Chemical Society, Division of Polymer Chemistry) (2010), 51(1), 12-13.

31-130-CAS-6995716	Steps: **1** Yield: **100%**
1.1	**Reagents:** Potassium hydroxide **Catalysts:** Palladium (peptide surface functionalized nanopa rticle), L-Threonyl-L-seryl-L-asparaginyl-L-alanyl-L-valyl-L-histidyl-L-prolyl-L-threonyl-L-leucyl-L-arginyl-L-histidyl-L-leucine (surface functionalized palladium nanoparticle with) **Solvents:** Water; 24 h, rt
1.2	**Reagents:** Hydrochloric acid **Solvents:** Water
Experimental Protocols	

Biomimetic Synthesis of Pd Nanocat alysts for the Stille Coupling Reaction

By: Pacardo, Dennis B.; et al

ACS Nano (2009), 3(5), 1288-1296.

31-130-CAS-13962126	Steps: **1** Yield: **100%**
1.1	**Reagents:** Sodium bicarbonate, Sodium hydroxide **Catalysts:** Palladium(1+), diaqua[2-[[bis(1-methylethyl)phosphino-κ*P*]oxy]-4-methylphenyl-κ*C*]-, (*SP*-4-2)-, sulfate (2:1) **Solvents:** Water; 720 min, pH 10.5, 100 °C
Experimental Protocols	

pH-Dependent C-C Coupling Reactions Catalyzed by Water-Soluble Palladacyclic Aqua Catalysts in Water

By: Ogo, Seiji; et al

Organometallics (2006), 25(2), 331-338.

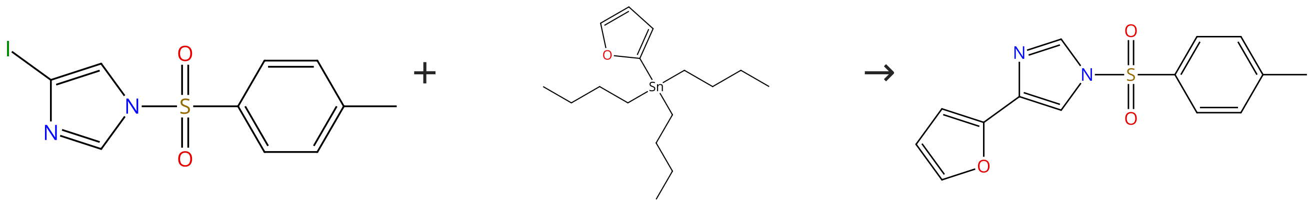
31-130-CAS-12576854	Steps: **1** Yield: **100%**
1.1	**Reagents:** Potassium hydroxide **Catalysts:** Palladium **Solvents:** Water; rt
1.2	**Reagents:** Hydrochloric acid **Solvents:** Water; rt
Experimental Protocols	

Dendrimer-Encapsulated Pd Nanoparticles as Aqueous, Room-Temperature Catalysts for the Stille Reaction

By: Garcia-Martinez, Joaquin C.; et al

Journal of the American Chemical Society (2005), 127(14), 5097-5103.

Scheme 27 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (36) Suppliers (45)

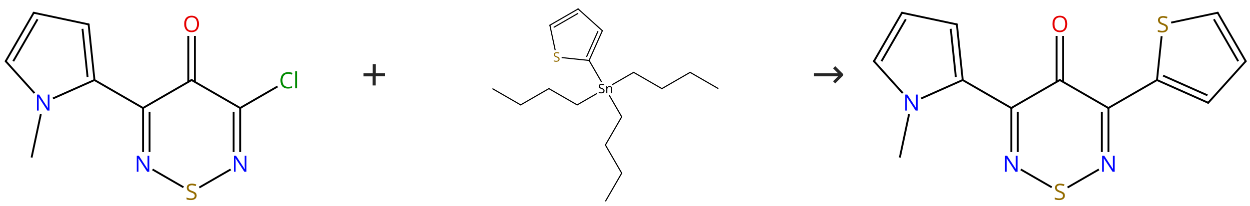
31-130-CAS-3309519	Steps: **1** Yield: **100%**
1.1	**Catalysts:** Tetrakis(triphenylphosphine)palladium **Solvents:** Dimethylformamide; 2 h, 110 °C

Stille Cross-Coupling for the Functionalization of the Imidazole Backbone: Revisit, Improvement, and Applications of the Method

By: Sandtorv, Alexander H.; et al

European Journal of Organic Chemistry (2015), 2015(16), 3506-3512.

Scheme 28 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (38)

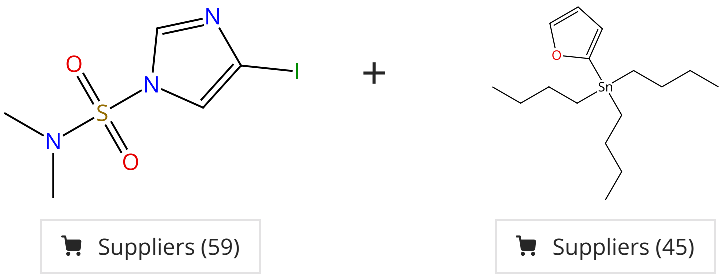
31-130-CAS-11368371	Steps: **1** Yield: **100%**
1.1	**Catalysts:** Dichlorobis(triphenylphosphine)palladium **Solvents:** Acetonitrile; rt; 45 min, 82 °C
Experimental Protocols	

Selective Stille Coupling Reactions of 3-Chloro-5-halo (pseudohalo)-4H-1,2,6-thiadiazin-4-ones

By: Ioannidou, Heraklidia A.; et al

Organic Letters (2011), 13(21), 5886-5889.

Scheme 29 (1 Reaction) Steps: **1** Yield: **100%**



31-130-CAS-15390210 Steps: **1** Yield: **100%**

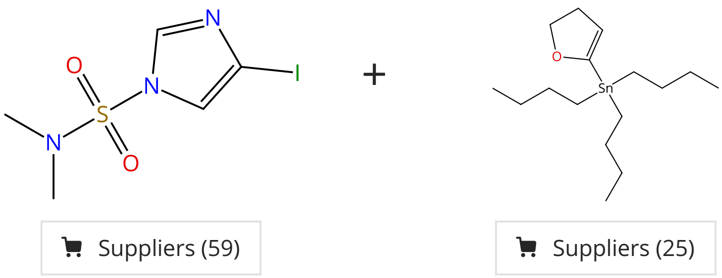
1.1 **Catalysts:** Cuprous iodide, Tetrakis(triphenylphosphine)palladium
Solvents: Dimethylformamide; 2 h, 110 °C

Stille Cross-Coupling for the Functionalization of the Imidazole Backbone: Revisit, Improvement, and Applications of the Method

By: Sandtorv, Alexander H.; et al

European Journal of Organic Chemistry (2015), 2015(16), 3506-3512.

Scheme 30 (1 Reaction) Steps: **1** Yield: **100%**



31-130-CAS-4411119 Steps: **1** Yield: **100%**

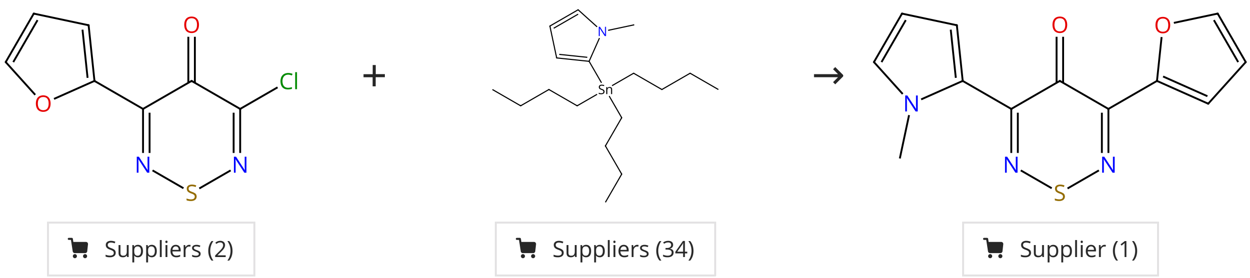
1.1 **Catalysts:** Cuprous iodide, Tetrakis(triphenylphosphine)palladium
Solvents: Dimethylformamide; 2 h, 110 °C

Stille Cross-Coupling for the Functionalization of the Imidazole Backbone: Revisit, Improvement, and Applications of the Method

By: Sandtorv, Alexander H.; et al

European Journal of Organic Chemistry (2015), 2015(16), 3506-3512.

Scheme 31 (1 Reaction) Steps: **1** Yield: **100%**



31-130-CAS-9259612 Steps: **1** Yield: **100%**

1.1 **Catalysts:** Dichlorobis(triphenylphosphine)palladium
Solvents: Acetonitrile; rt; 20 min, 82 °C

Experimental Protocols

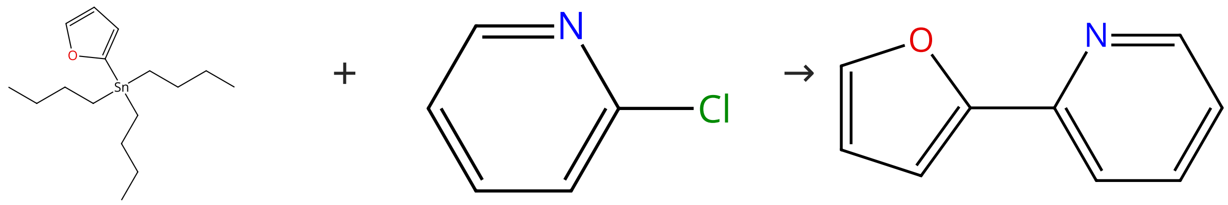
Selective Stille Coupling Reactions of 3-Chloro-5-halo(pseudohalo)-4H-1,2,6-thiadiazin-4-ones

By: Ioannidou, Heraklidia A.; et al

Organic Letters (2011), 13(21), 5886-5889.

Scheme 32 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (45) Suppliers (66) Suppliers (32)

31-130-CAS-13696432 Steps: **1** Yield: **100%**

1.1 **Reagents:** Lithium chloride
Catalysts: Dichlorobis(triphenylphosphine)palladium
Solvents: Tetrahydrofuran; 480 min, 85 °C

Experimental Protocols

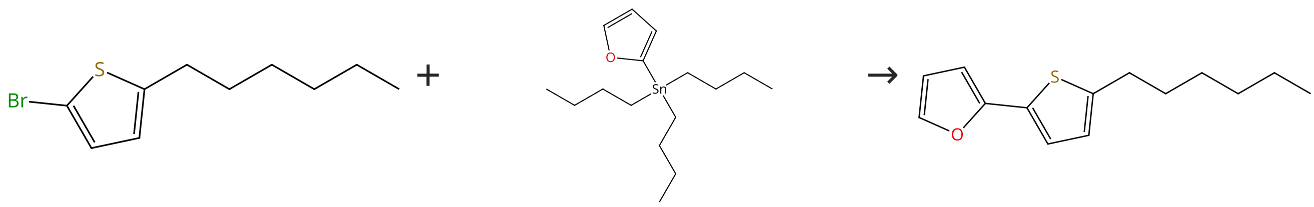
Dual Effect of Halides in the Stille Reaction: In Situ Halide Metathesis and Catalyst Stabilization

By: Verbeeck, Stefan; et al

Chemistry - A European Journal (2010), 16(43), 12831-12837, S12831/1-S12831/17.

Scheme 33 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (67) Suppliers (45)

31-130-CAS-6588726 Steps: **1** Yield: **100%**

1.1 **Reagents:** Copper oxide (CuO)
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Dimethylformamide; 5 min, 100 °C; 1 h, 100 °C; 100 °C → rt

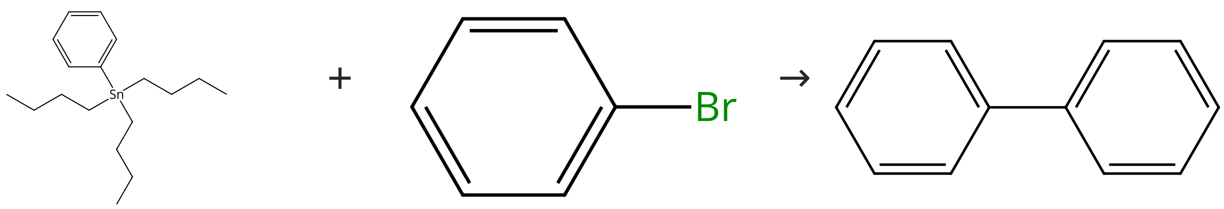
Synthesis of oligo(thienylfuran)s with thiophene rings at both ends and their structural, electronic, and field-effect properties

By: Miyata, Yasuo; et al

Chemistry - An Asian Journal (2007), 2(12), 1492-1504.

Scheme 34 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (41) Suppliers (81) Suppliers (116)

31-130-CAS-5462332 Steps: **1** Yield: **100%**

1.1 **Reagents:** Potassium fluoride
Catalysts: Copper oxide (Cu_2O), Tri-*o*-tolylphosphine
Solvents: Tetrabutylammonium bromide; 28 h, 125 - 130 °C

Experimental Protocols

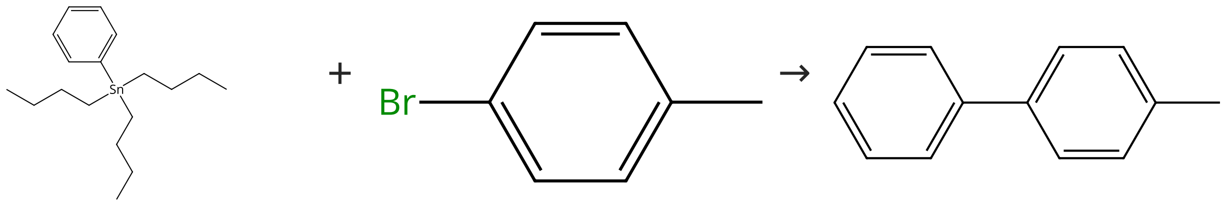
Reusable Copper-Catalyzed Cross-Coupling Reactions of Aryl Halides with Organotins in Inexpensive Ionic Liquids

By: Li, Jin-Heng; et al

Journal of Organic Chemistry (2006), 71(19), 7488-7490.

Scheme 35 (2 Reactions)

Steps: **1** Yield: **100%**



Suppliers (41) Suppliers (67) Suppliers (82)

31-130-CAS-1616802

Steps: **1** Yield: **100%**

1.1 **Reagents:** (+)-Lactose, Dimethylurea, Ammonium chloride
Catalysts: Triphenylarsine, Palladium, tris[μ-[(1,2-η:4,5-η)-(1*E*,4*E*)-1,5-diphenyl-1,4-pentadien-3-one]]di-, compd. with trichloromethane (1:1); 6 h, 90 °C; 90 °C → rt

Stille reactions with tetraalkylstannanes and phenyltrialkylstannanes in low melting sugar-urea-salt mixtures

By: Imperato, Giovanni; et al

Advanced Synthesis & Catalysis (2006), 348(15), 2243-2247.

31-130-CAS-2565944

Steps: **1** Yield: **100%**

1.1 **Reagents:** Potassium hydroxide
Catalysts: Dabco, Palladium diacetate, Polyethylene glycol
Solvents: Water; 15 h, 80 °C

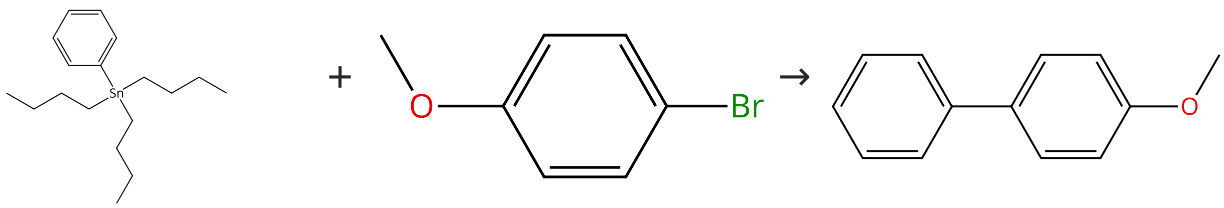
PEG-400 promoted Pd(OAc)$_2$/DABCO-catalyzed cross-coupling reactions in aqueous media

By: Li, Jin-Heng; et al

Tetrahedron (2006), 62(1), 31-38.

Scheme 36 (3 Reactions)

Steps: **1** Yield: **100%**



Suppliers (41) Suppliers (71) Suppliers (69)

31-614-CAS-38361614

Steps: **1** Yield: **100%**

1.1 **Reagents:** Cesium fluoride
Catalysts: Bis[2'-(amino-κ*N*)[1,1'-biphenyl]-2-yl-κ*C*]bis[μ-(methanesulfonato-κ*O*:κ*O*)]dipalladium, 3016418-42-8
Solvents: 2-Methyltetrahydrofuran; 6 h, 80 °C

Experimental Protocols

Bulky, electron-rich, renewable: analogues of Beller's phosphine for cross-couplings

By: van der Westhuizen, Danielle; et al

Catalysis Science & Technology (2023), 13(23), 6733-6742.

31-614-CAS-38361659

Steps: **1** Yield: **100%**

1.1 **Reagents:** Cesium fluoride
Catalysts: Bis[2'-(amino-κ*N*)[1,1'-biphenyl]-2-yl-κ*C*]bis[μ-(methanesulfonato-κ*O*:κ*O*)]dipalladium, Methanesulfonic acid, 1,1,1-trifluoro-, compd. with bis(1,1,5-trimethylheptyl) phosphine (1:1)
Solvents: 2-Methyltetrahydrofuran; 6 h, 80 °C

Experimental Protocols

Bulky, electron-rich, renewable: analogues of Beller's phosphine for cross-couplings

By: van der Westhuizen, Danielle; et al

Catalysis Science & Technology (2023), 13(23), 6733-6742.

31-130-CAS-7705138

Steps: **1** Yield: **100%**

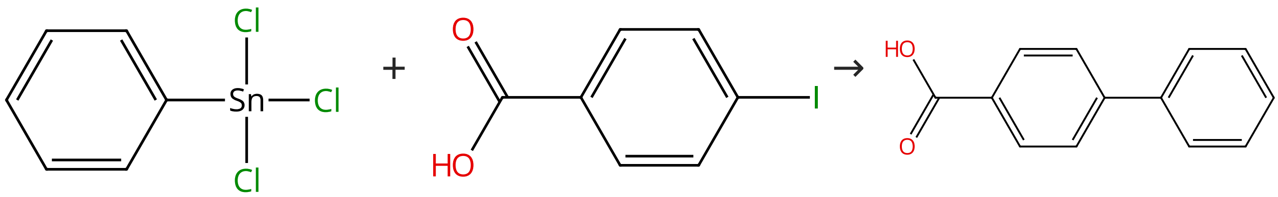
1.1 **Reagents:** (+)-Lactose, Dimethylurea, Ammonium chloride
Catalysts: Triphenylarsine, Palladium, tris[μ-[(1,2-η:4,5-η)-(1*E*,4*E*)-1,5-diphenyl-1,4-pentadien-3-one]]di-, compd. with trichloromethane (1:1); 6 h, 90 °C; 90 °C → rt

Stille reactions with tetraalkylstannanes and phenyltrialkylstannanes in low melting sugar-urea-salt mixtures

By: Imperato, Giovanni; et al

Advanced Synthesis & Catalysis (2006), 348(15), 2243-2247.

Scheme 37 (6 Reactions) Steps: **1** Yield: **100%**



Suppliers (51) Suppliers (94) Suppliers (101)

31-130-CAS-4627098 Steps: **1** Yield: **100%**

1.1 **Reagents:** Potassium hydroxide
Catalysts: Palladium, L-Threonyl-L-seryl-L-asparaginyl-L-alanyl-L-valyl-L-histidyl-L-prolyl-L-threonyl-L-leucyl-L-arginyl-L-histidyl-L-leucine (palladium nanoparticle surface functionalized with)
Solvents: Water; 24 h, rt
1.2 rt

Interrogating the catalytic mechanism of nanoparticle mediated Stille coupling reactions employing bio-inspired Pd nanocatalysts

By: Pacardo, Dennis B.; et al

Nanoscale (2011), 3(5), 2194-2201.

31-130-CAS-14392741 Steps: **1** Yield: **100%**

1.1 **Reagents:** Potassium hydroxide
Solvents: Water; rt
1.2 **Reagents:** Oxygen
Catalysts: Palladium; 24 h, rt
1.3 **Reagents:** Hydrochloric acid
Solvents: Water; rt

Experimental Protocols

Effects of the Material Structure on the Catalytic Activity of Peptide-Templated Pd Nanomaterials

By: Bhandari, Rohit; et al

ACS Catalysis (2011), 1(2), 89-98.

31-130-CAS-9322109 Steps: **1** Yield: **100%**

1.1 **Reagents:** Potassium hydroxide
Catalysts: Palladium
Solvents: Water; 24 h, 25 °C
1.2 **Reagents:** Hydrochloric acid
Solvents: Water

Exploring bio-inspired strategies for the production of noble metal nanocatalysts

By: Knecht, Marc R.; et al

Polymer Preprints (American Chemical Society, Division of Polymer Chemistry) (2010), 51(1), 12-13.

31-130-CAS-2739940 Steps: **1** Yield: **100%**

1.1 **Reagents:** Potassium hydroxide
Catalysts: Palladium (peptide surface functionalized nanoparticle), L-Threonyl-L-seryl-L-asparaginyl-L-alanyl-L-valyl-L-histidyl-L-prolyl-L-threonyl-L-leucyl-L-arginyl-L-histidyl-L-leucine (surface functionalized palladium nanoparticle with)
Solvents: Water; 24 h, rt
1.2 **Reagents:** Hydrochloric acid
Solvents: Water

Experimental Protocols

Biomimetic Synthesis of Pd Nanocatalysts for the Stille Coupling Reaction

By: Pacardo, Dennis B.; et al

ACS Nano (2009), 3(5), 1288-1296.

31-130-CAS-1561713 Steps: **1** Yield: **100%**

1.1 **Reagents:** Potassium hydroxide
Catalysts: Palladium, Ethylenediamine-methyl acrylate copolymer
Solvents: Water; 24 h, 40 °C

Experimental Protocols

Dendrimer-Encapsulated Pd Nanoparticles versus Palladium Acetate as Catalytic Precursors in the Stille Reaction in Water

By: Bernechea, Maria; et al

Inorganic Chemistry (2009), 48(10), 4491-4496.

31-130-CAS-3170646 Steps: **1** Yield: **100%**

1.1 **Reagents:** Potassium hydroxide
Catalysts: Palladium
Solvents: Water; 15 h, rt

1.2 **Reagents:** Hydrochloric acid
Solvents: Water; rt

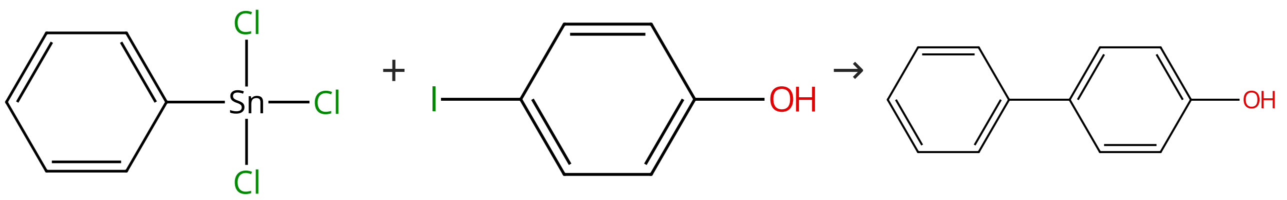
Experimental Protocols

Dendrimer-Encapsulated Pd Nanoparticles as Aqueous, Room-Temperature Catalysts for the Stille Reaction

By: Garcia-Martinez, Joaquin C.; et al

Journal of the American Chemical Society (2005), 127(14), 5097-5103.

Scheme 38 (2 Reactions) Steps: **1** Yield: **100%**



Suppliers (51) Suppliers (86) Suppliers (105)

31-130-CAS-6548266 Steps: **1** Yield: **100%**

1.1 **Reagents:** Potassium hydroxide
Catalysts: Palladium
Solvents: Water; 24 h, 25 °C

1.2 **Reagents:** Hydrochloric acid
Solvents: Water

Exploring bio-inspired strategies for the production of noble metal nanocatalysts

By: Knecht, Marc R.; et al

Polymer Preprints (American Chemical Society, Division of Polymer Chemistry) (2010), 51(1), 12-13.

31-130-CAS-9137194 Steps: **1** Yield: **100%**

1.1 **Reagents:** Potassium hydroxide
Catalysts: Palladium (peptide surface functionalized nanoparticle), L-Threonyl-L-seryl-L-asparaginyl-L-alanyl-L-valyl-L-histidyl-L-prolyl-L-threonyl-L-leucyl-L-arginyl-L-histidyl-L-leucine (surface functionalized palladium nanoparticle with)
Solvents: Water; 24 h, rt

1.2 **Reagents:** Hydrochloric acid
Solvents: Water

Experimental Protocols

Biomimetic Synthesis of Pd Nanocat alysts for the Stille Coupling Reaction

By: Pacardo, Dennis B.; et al

ACS Nano (2009), 3(5), 1288-1296.